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Radiography and imaging sciences (Kenya Medical Training College)

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KENYA MEDICAL TRAINING COLLEGE
FACULTY OF DIAGNOSTIC SCIENCES
DEPARTMENT OF RADIOGRAPHY AND IMAGING

FINAL QUALIFYING EXAMINATION
FOR
DIPLOMA IN RADIOGRAPHY AND IMAGING
PAPER: IMAGE PROCESSING TECHNOLOGY

DATE: 16th JANUARY, 2024 **TIME: 3 HOURS (9:00 – 12:00 NOON)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (40 Multiple Choice Questions)
   - Section 2 (8 Short Answer Questions)
   - Section 3 (1 Long Answer Question)
2. Attempt **ALL** Questions
3. ALL Multiple choice questions MUST be answered in **CAPITAL LETTERS**
4. Write your **EXAMINATION NUMBER** on all the answer sheets provided and on the question paper.
5. No other writing should be done on the question paper except your Examination Number.
6. Ensure that all examination answer scripts are handed in at the end of the examination
7. Ensure you sign the examination register provided

EXAMINATION NUMBER ....D/MIS/JA.....

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SECTION 1: MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. Which is the correct order of steps in the traditional film processing?
   A. Fixing, developing, washing, drying
   B. Developing, fixing, washing, drying
   C. Washing, developing, fixing, drying
   D. Drying, washing, developing, fixing
2. What is the purpose of developer solution in traditional film processing is to?
   A. Remove unexposed silver latitude crystals
   B. Reduce exposed silver latitude crystals to metallic silver
   C. Harden the emulsion layer
   D. Neutralize any remaining chemicals in the film
3. Which of the following is a disadvantage of traditional radiographic film compared to digital radiography?
   A. Limited dynamic range
   B. High cost of equipment
   C. Poor spatial resolution
   D. Limited availability of manipulation tools
4. The main purpose of intensifying screen in traditional radiography is to_____
   A. Reduce radiation exposure to patients
   B. Increase radiation exposure to patients
   C. Increase the speed of the film
   D. Decrease the contrast of the image
5. What is the commonly used phosphor material in intensifying screens?
   A. Calcium tungstate
   B. Aluminum Oxide
   C. Lead oxide
   D. Barium Sulphate
6. What is the relationship between the size of phosphor crystals in an intensifying screen and spatial resolution of the image?
   A. Directly proportional
   B. Inversely proportional
   C. No relationship
   D. Depends of the type of film used
7. Which of the following is a disadvantage of using intensifying screens in radiography?
   A. Reduced image contrast
   B. Increased patient radiation dose
   C. Decreased image speed
   D. Increased image resolution

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8. A group of students were discussing the primary purpose of a radiographic cassette in traditional radiography. What would be the best choice?
   A. Hold the intensifying screens and films in place
   B. Reduce radiation exposure to the patient
   C. Increase radiation exposure to the patient
   D. Increase the spatial resolution to the image
9. What is the material commonly used in the construction of radiographic cassettes?
   A. Aluminium
   B. Lead
   C. Copper
   D. Steel
10. A department is in the process of procuring traditional radiography cassettes. Which of the following closure type would you recommend?
   A. Latch-type closure
   B. Hinged closure
   C. Snap-type closure
   D. Magnetic closure
11. Which of the following factor can affect the image quality in traditional radiography?
   A. Type of film used
   B. Size of the cassette
   C. Type of cassette closure
   D. Thickness of the cassette
12. Image acquisition in computed radiography follows specific steps. Which of the following is the correct order?
   A. Exposure, scan, processing, display
   B. Processing, scan, exposure, display
   C. Scan, exposure, processing, display
   D. Exposure, processing, scan, display
13. The main purpose of the photostimulable phosphor plate in computed radiography is _____
   A. Reduce patient radiation dose
   B. Increase the speed of image acquisition
   C. Capture the X-ray image
   D. Convert X-rays to visible light
14. Which of the following is a disadvantage of computed radiography compared to digital radiography?
   A. Longer processing time
   B. Higher patient radiation dose
   C. Limited dynamic range
   D. Lower spatial resolution

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15. Which of the following is a benefit of using wireless digital detector in digital radiography?
    A. Faster image acquisition time
    B. Higher spatial resolution
    C. Lower patient radiation dose
    D. Greater dynamic range
16. The purpose of image post-processing in digital radiography is to_____
    A. Increase the speed of image acquisition
    B. Reduce patient's radiation dose
    C. Improve image quality
    D. Limit the amount of scatter radiation
17. Which type of digital radiography utilizes a flat panel detector to capture the X-ray image?
    A. Direct digital radiography
    B. Indirect digital radiography
    C. Computed radiography
    D. Film-screen radiography
18. Students were criticizing digital radiography over traditional film-screen radiography. Which of the following statements supports the argument?
    A. Lower patient radiation dose
    B. Lower spatial resolution
    C. Limited dynamic range
    D. Longer image acquisition time
19. Which of the following is the purpose of digital image processing software in digital radiography?
    A. Capture the X-ray image
    B. Adjust image brightness and contrast
    C. Reduce the patient's radiation dose
    D. Convert X-rays to visible light
20. Identify the advantage of digital radiography over computed radiography (CT)?
    A. Higher spatial resolution
    B. Greater dynamic range
    C. Lower patient radiation dose
    D. Ability to provide cross-sectional images
21. Radiography students were discussing the concept of Sensitometry. Which of the following statements best describe it?
    A. Formation of image on the image receptor
    B. The effect of X-rays on the image receptor
    C. The response of image receptor to radiation
    D. The interpretation of image by radiologist

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22. Which of the following measures the speed of an X-ray film?
    A. Density
    B. Contrast
    C. Gamma
    D. Sensitivity
23. _____ measures the range of exposures that a radiographic film can record.
    A. Contrast
    B. Density
    C. Latitude
    D. Sensitivity
24. What is measured by the characteristic curve in radiographic sensitometry?
    A. Speed
    B. Contrast
    C. Resolution
    D. Spectral sensitivity
25. A chest radiograph with high density was presented to a group of students for discussion. What is the relationship between the film density and X-ray exposure?
    A. Film density increases with increase in X-ray exposure
    B. Film density decreases with increase in X-ray exposure
    C. Film density is not affected by X-ray exposure
    D. Film density increases with X-ray exposure
26. An image quality indicator (IQI) in radiography is used to measure _____.
    A. Patient radiation dose
    B. Image contrast
    C. Image resolution
    D. Image sharpness
27. What is the relationship between X-ray beam collimation and radiographic image quality?
    A. Increased collimation, increased image contrast
    B. Decreased collimation, decreased image resolution
    C. Decreased collimation, decreased patient radiation dose
    D. Increased collimation, decreased image noise
28. Which of the following is the main purpose of an anti-scatter grid in radiography?
    A. Reduce image noise
    B. Improve spatial resolution
    C. Increase image contrast
    D. Reduce patient radiation dose
29. Which of the following is the key measure of X-ray beam quality?
    A. Kilovoltage peak (kVp)
    B. Milliamperage (mA)
    C. Exposure Time
    D. Source-to-image distance (SID)

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30. What is measured by the Modulation Transfer Function (MTF) in radiographic image quality?
    A. Contrast
    B. Noise
    C. Spatial resolution
    D. Dynamic range
31. A radiographer presented a radiographic image with an artifact to students during clinical rotation. Which of the following best defines a radiographic image artifact?
    A. An object or structure imaged by accident
    B. An image feature that is not representative of the image being imaged
    C. A measurement error in the X-ray beam
    D. A defect in the radiographic film detector
32. Motion artifacts in radiographic images can be caused by_____.
    A. Patient immobilization
    B. Equipment vibration
    C. Poor film-screen contact
    D. High exposures
33. Digital radiography has many image artifacts, Moire Pattern Artifact is an example. What is the main cause of this artifact?
    A. Inadequate X-ray beam penetration
    B. Insufficient detector sampling frequency
    C. Poor image processing algorithms
    D. Patient motion during exposure
34. The role of image intensifier in fluoroscopy is to _____.
    A. Absorb and attenuate X-rays
    B. Amplify and convert X-rays into visible light
    C. Filter and collimate the X-ray beam
    D. Generate and control the X-ray beam
35. What is the purpose of breast compression in mammography imaging?
    A. Reduce patient discomfort
    B. Improve image quality and reduce radiation dose
    C. Detect breast abnormalities
    D. Monitor breast cancer treatment
36. What is the primary modality used in mammography?
    A. X-ray
    B. Ultrasound
    C. MRI
    D. CT
37. Which of the following is used in Digital mammography?
    A. A film based system.
    B. A digital detector.
    C. Magnetic Resonance Imaging.
    D. Ultrasound.

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38. The primary goal of quality assurance in mammography is to ensure that_____.
    A. Machines are properly calibrated and maintained
    B. Images are accurately interpreted and diagnosed
    C. Patients receive comfortable and timely mammograms
    D. Radiation to the patient is minimized
39. What is the purpose of Reconstruction algorithms in CT scan?
    A. Improve spatial resolution and contrast
    B. Reduce radiation dose to the patient
    C. Make the CT scan procedure more comfortable
    D. Decrease the time it takes to acquire images
40. A clinician requested for a repeat of a CT scan of the abdomen with contrast media. What does the contrast agent aid in?
    A. Improving the visualization and characterization of certain tissues and structures.
    B. Reducing radiation dose to the patients.
    C. Make the CT scan procedure more comfortable to the patient.
    D. Decrease the time it takes to acquire the CT scan image.

SECTION 2: SHORT ANSWER QUESTIONS (40 MARKS)

41. Describe the modulation transfer function (MTF) in radiographic image quality (5 marks)
42. State five (5) major development in Radiographic Image Progressing Technology (5 marks)
43. Elucidate five (5) advantages of using digital detector in fluoroscopy (5 marks)
44. Enumerate five (5) ways in which digital radiography has improved image recording (5 marks)
45. Explain how sensitometry relates to image quality in radiography (5 marks)
46. Describe the operating principle of Computed Radiography (CR) (6 marks)
47. Differentiate between spatial resolution and contrast resolution in radiographic imaging and give two examples in each case (4 marks)
48. State five parameters commonly used to describe radiographic image quality (5 marks)

SECTION 3: LONG ANSWER QUESTIONS (20 MARKS)

49. a) Compare computed radiography (CR) and direct digital radiography (DR) (10 marks)
    b) Explain five (5) common digital image post processing procedures (10 marks)

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